FYS4480/9480 November 6

FYS4480/9480 November 6

Diagram rules:

2nd (2p2h)

$$\frac{1}{4}\sum_{\substack{ij\\ab}} \frac{\bar{v}_{ij}^{ab}\bar{v}_{ab}^{ij}}{\varepsilon_i+\varepsilon_j-\varepsilon_a-\varepsilon_b}$$

single-particle basis

$\langle \Phi_b | H_I R_0 H_I | \Phi_c \rangle$

SD basis

$V_{cm} = \langle \Phi_c | H_1 | \Phi_m \rangle$

$m \neq 0$

$$\Delta E^{(2)} = \sum_{m>1} \frac{V_{om} V_{mo}}{\varepsilon_o - \varepsilon_m}$$

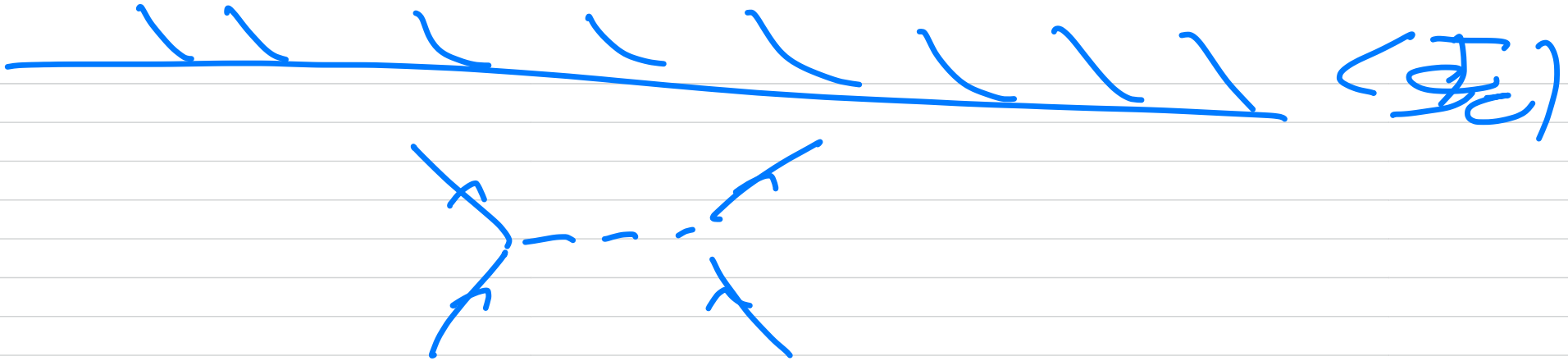
$$m \in \{2p\,2s\}$$

$$m \in \{1p\,1s\}$$

sp-basis:

$$\sum_{ia} \frac{\tilde{f}_{ia}\,\tilde{f}_{ai}}{\varepsilon_{i'} - \varepsilon_a}$$

$$\tilde{f}_{ia} = \sum_{j \le F} \bar{v}_{ij'aj'}$$



$$R_0 \sum_m \frac{|\Phi_m\rangle\langle\Phi_m|}{E_0 - E_m}$$

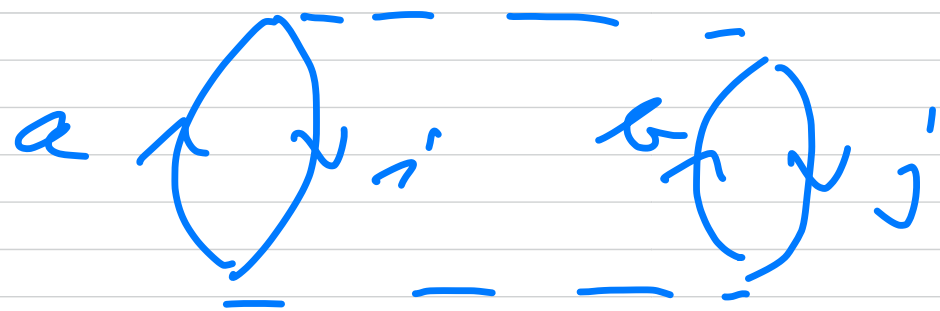
$\langle \Phi_m | H_1 | \Phi$

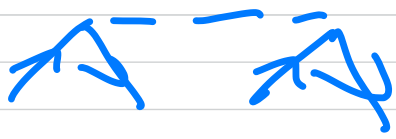
V

$|p|_k$

$|2p\,2_k\rangle$

$|\Phi_c\rangle$

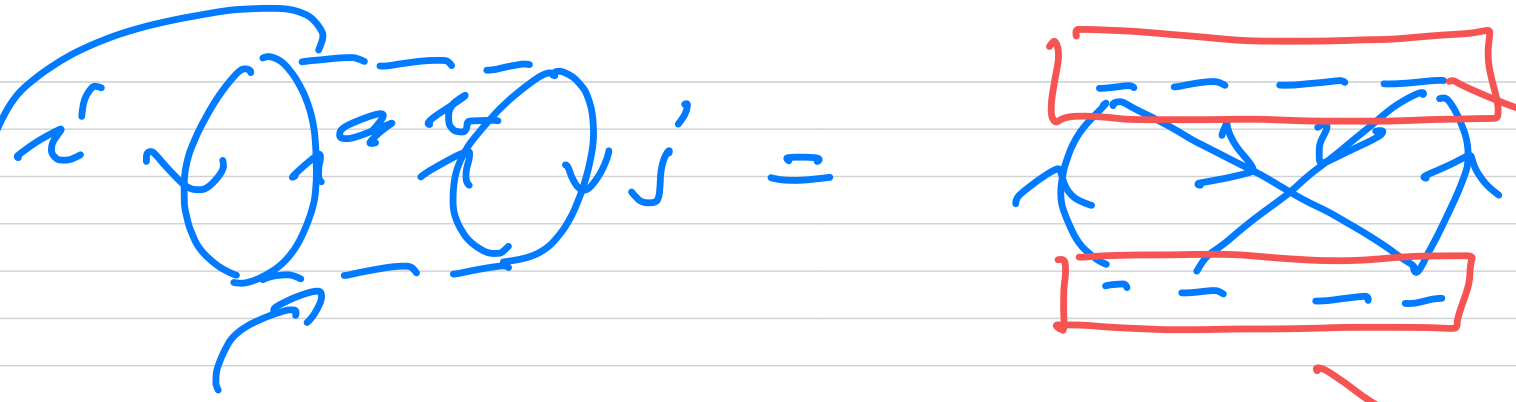






$$= \frac{1}{4} \sum_{\substack{ab \\ ij}} \frac{\bar{v}_{ij}^{ab}\, \bar{v}_{ab}^{ij}}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}$$

(i) Draw all topologically distinct diagrams to a given order given by the symmetries of the interaction.



$$\bar{v}_{ab,i'j'} =$$

$$\langle ab|v|i'j'\rangle_{AS} = -\langle ab|v|j'i'\rangle_{AS}$$

$$\rightarrow \langle i'j'|v|ab\rangle_{AS} = -\langle ab|v|j'i'\rangle_{AS}$$

$$\sum_{\substack{ab \\ i'j'}} \frac{\bar{v}_{i'j',ab}\,\bar{v}_{ab,i'j'}}{\varepsilon_{i'}+\varepsilon_{j'}-\varepsilon_a-\varepsilon_b}$$

(ii) Label all particle lines with abc... and upgoing arrow.
Label all hole lines with ijk... and downgoing arrows

(iii) Sum freely over all internal lines between vertices

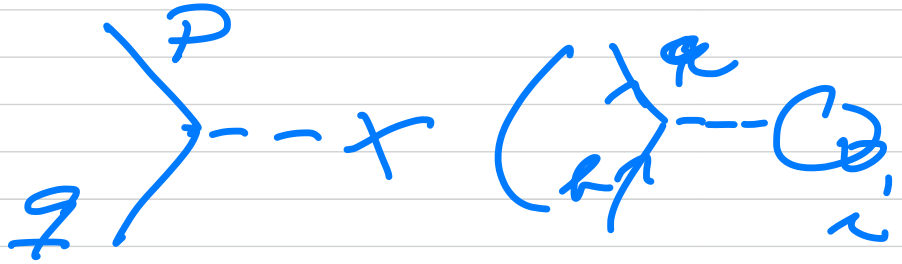
(iv) Assign interaction vertices. For each

vertex we have a pair (p, q) where p is an outgoing sp state and q is an incoming sp state

a two-body vertex

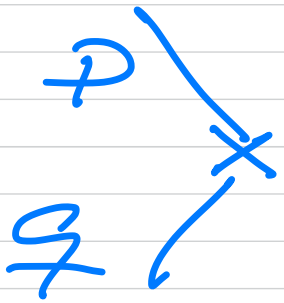


a one-body



$$\sum_{i \in F} \overline{v_{qi}} v_{pi}$$

or



(V) Between interaction vertices, assign a factor

$$\frac{1}{\sum_i \varepsilon_i' - \sum_a \varepsilon_a}$$

(from R_0)

$\sum_i$: sum over intermediate hole states

$\sum_a$: sum over intermediate particle states

(vi) assign a factor

$$(-1)^{n_e + n_h}$$

n_h = # hole lines

n_e = # closed loops

(vii) assign a factor

$$\left(\frac{1}{2}\right)^{n_{eqp}}$$

n_{eqp} = # of equivalent pairs

equivalent pairs are defined
at two sp states (Both
holes or both particles)
which start and end
at the same vertex

Example

2p2h

$$\bar{v}_{ij'ab} \left(\quad \right) - \bar{v}_{abij'}$$

$$n_{eqp} = 2 \Rightarrow \left(\frac{1}{2}\right)^2 = 1/4$$

$$\left.\begin{aligned} n_h &= 2 \\ n_e &= 2 \end{aligned}\right\} (-1)^4 = +1$$

Denominator: $\dfrac{1}{\varepsilon_{i'} + \varepsilon_{j'} - \varepsilon_a - \varepsilon_b}$

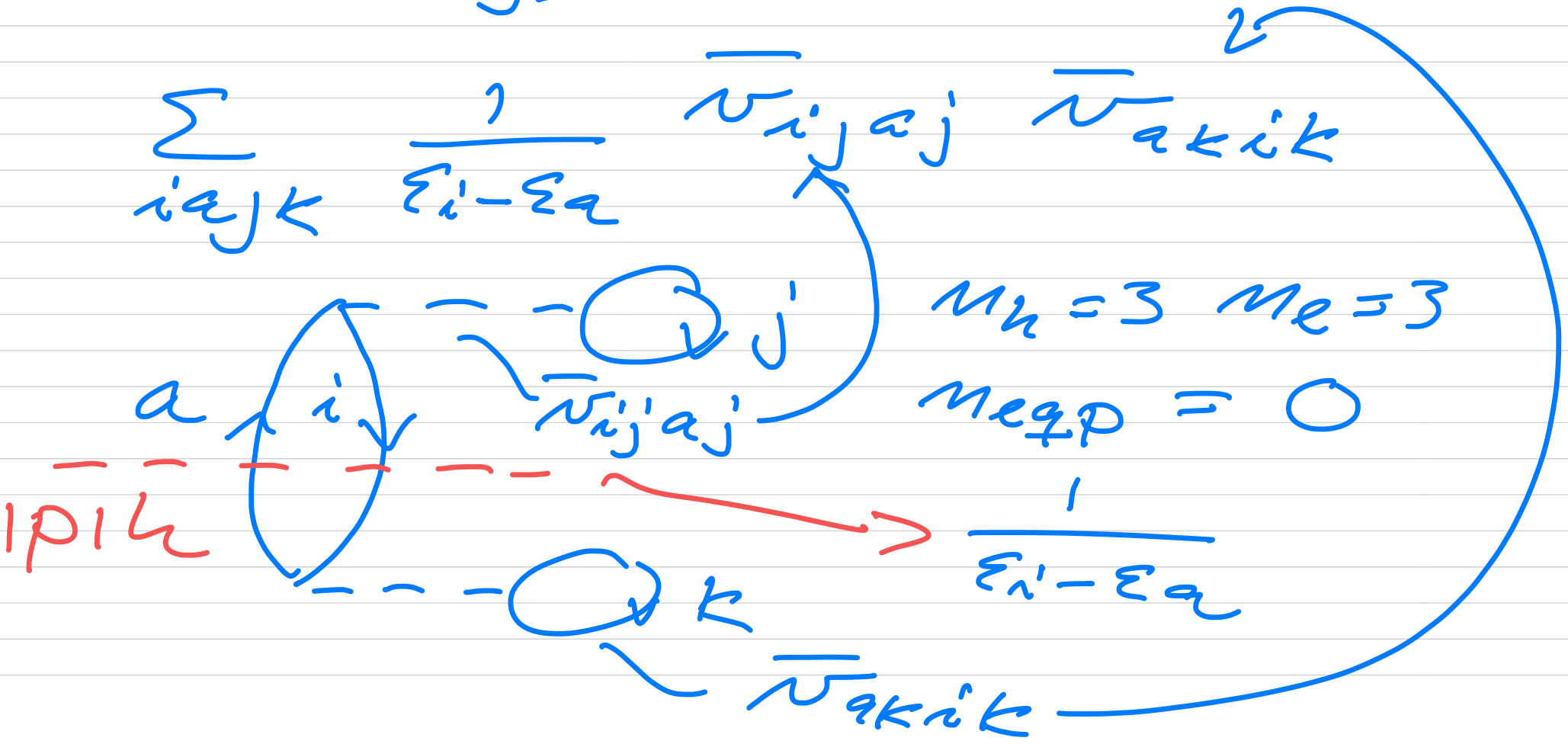
$$\tilde{v} + \frac{1}{4} \sum_{\substack{ab \\ i'j'}} \frac{\bar{v}_{ij'ab}\, \bar{v}_{abij'}}{\varepsilon_{i'} + \varepsilon_{j'} - \varepsilon_a - \varepsilon_b}$$

(2p2h)

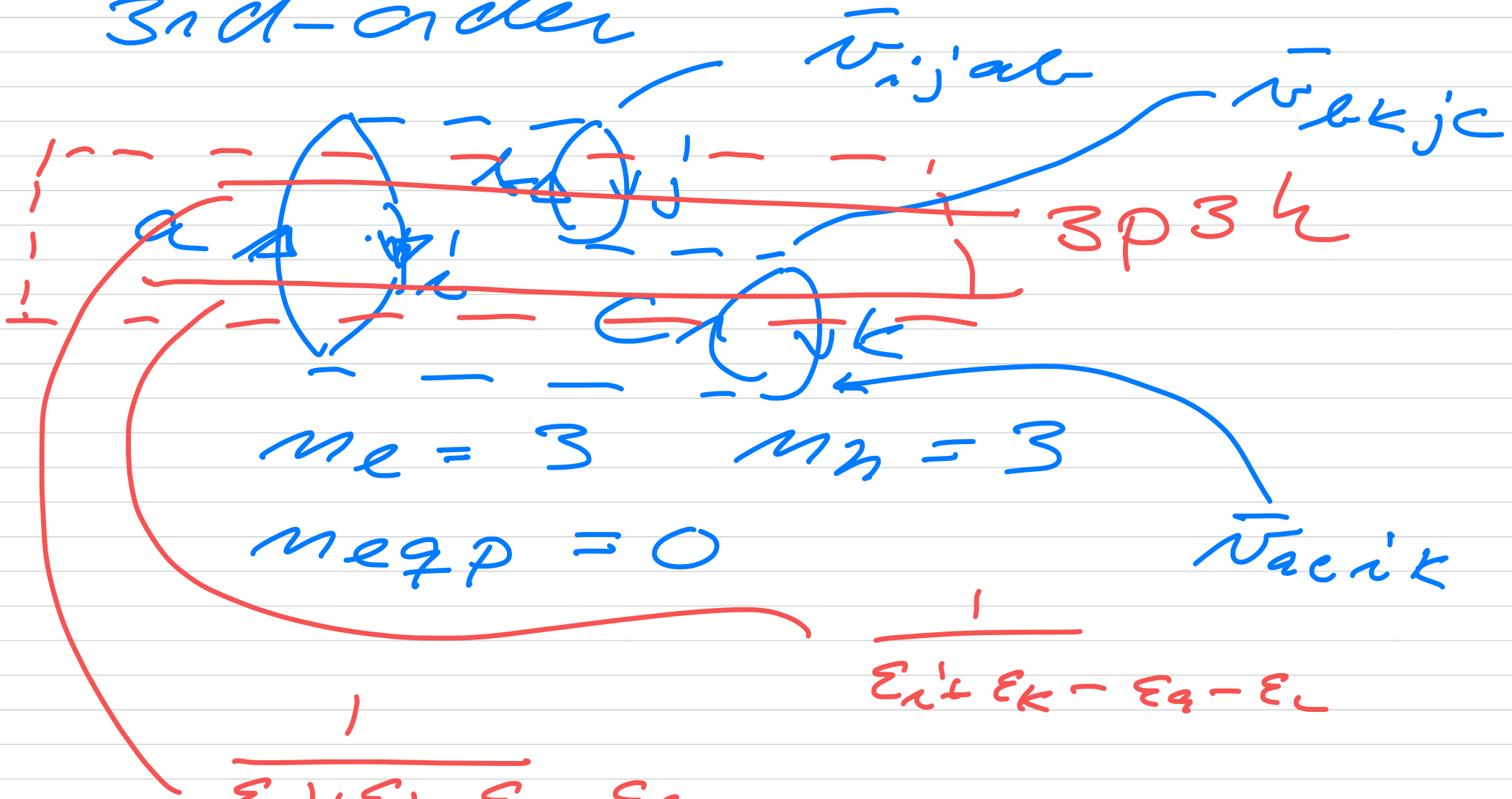
$$|pl\rangle : \sum_{ia} \frac{\tilde{f}_{ia} f_{ai'}}{\varepsilon_{i'} - \varepsilon_a}$$

$$\tilde{f}_{ia} = \sum_{j \le F} \bar{v}_{ij a_j'}$$

$$\sum_{iajk} \frac{1}{\varepsilon_{i'} - \varepsilon_a} \bar{v}_{ij a_j} \bar{v}_{akik}$$

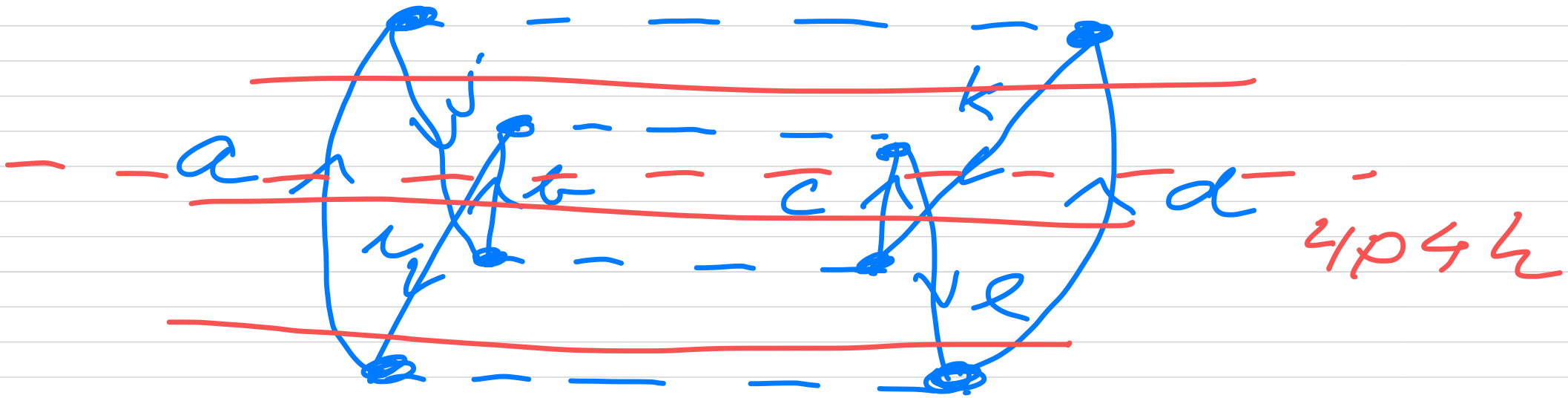


3rd-order



$$+1 \times \sum_{\substack{abc \\ i'jk}} \frac{\bar{v}_{ij'ab}\,\bar{v}_{bkjc}\,\bar{v}_{ac i'k}}{(\varepsilon_{i'} + \varepsilon_k - \varepsilon_a - \varepsilon_c)(\varepsilon_{i'} + \varepsilon_{j'} - \varepsilon_a - \varepsilon_b)}$$

4-th order

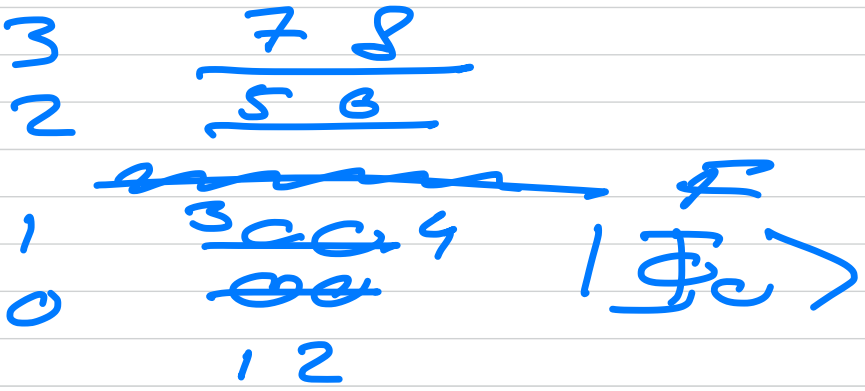


$n_e = 0 \qquad n_h = 4 \qquad +1$

$n_{eqp} = 4$

$$\frac{1}{16} \sum_{\substack{abcd \\ ijkl}} \frac{\bar{v}_{jkab}\, \bar{v}_{ilbc}\, \bar{v}_{bcjk}\, \bar{v}_{adil}}{(\varepsilon_j + \varepsilon_k - \varepsilon_a - \varepsilon_b)(\varepsilon_i + \varepsilon_j + \varepsilon_k + \varepsilon_l - \varepsilon_a - \varepsilon_b - \varepsilon_c - \varepsilon_d)} \times \frac{1}{\varepsilon_i + \varepsilon_l - \varepsilon_a - \varepsilon_d}$$

Pairing example (1st Martin)



$|\Phi_0\rangle$

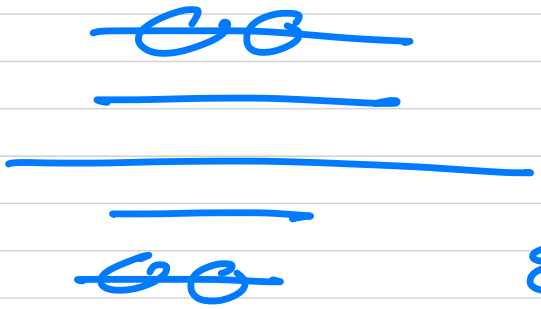
$\varepsilon_0 = 2 \cdot \alpha = 2$

$a = 1$



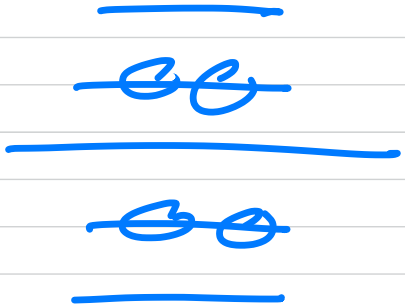
$|\Phi_1\rangle$

$\varepsilon_1 = 4$



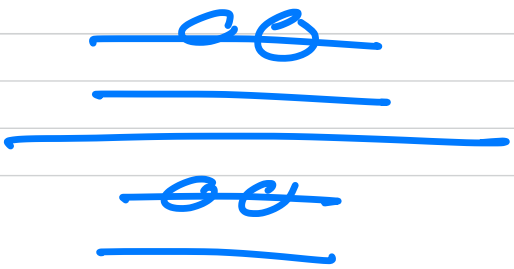
$|\Phi_2\rangle$

$\varepsilon_2 = 6$



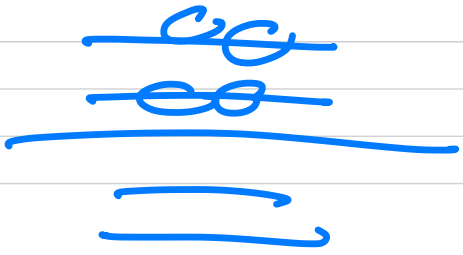
$|\Phi_3\rangle$

$\varepsilon_3 = 6$



$|\Phi_4\rangle$

$\varepsilon_4 = 8$



$|\Phi_5\rangle$

$\varepsilon_5 = 10$

$$\sum_m \frac{\langle \Phi_c | H_1 | \Phi_m \rangle \langle \Phi_m | H_1 | \Phi_c \rangle}{E_0 - E_m}$$

$$\langle \Phi_c | H_1 | \Phi_S \rangle = 0$$

occ vir

$$H = \begin{bmatrix} 2-g & -g/2 & -g/2 & -g/2 & -g/2 & 0 \\ -g/2 & 4-g & - & - & & \\ -g/2 & & 6-g & - & & \\ & & & 6-g & - & \\ & & & & 8-g & \\ & & & & & 10-g \end{bmatrix}$$

$$\langle \Phi_c | H_1 | \Phi_m \rangle = -g/2$$

except for $m=0$ or $m=$ then zero

$$\Delta E^{(2)} = g^2/4 \left[\frac{1}{2-4} + \frac{1}{2-6} + \frac{1}{2-6} + \frac{1}{2-8} \right]$$

$$= \boxed{-\,5^2\,7/24}$$

$$\Delta E^{(2)} = \langle \Phi_0 | H_1 R_0 H_1 | \Phi_0 \rangle$$

$$\langle \Phi_0 | H_1 | \Phi_m \rangle =$$

$$V_{0m} = [\,0 \quad -3/2 \quad -5/2 \quad -5/2 \quad -3/2\,]$$

Φ_5

$$R_0 = \sum_m \frac{|\Phi_m\rangle\langle\Phi_m|}{\varepsilon_0 - \varepsilon_m}$$

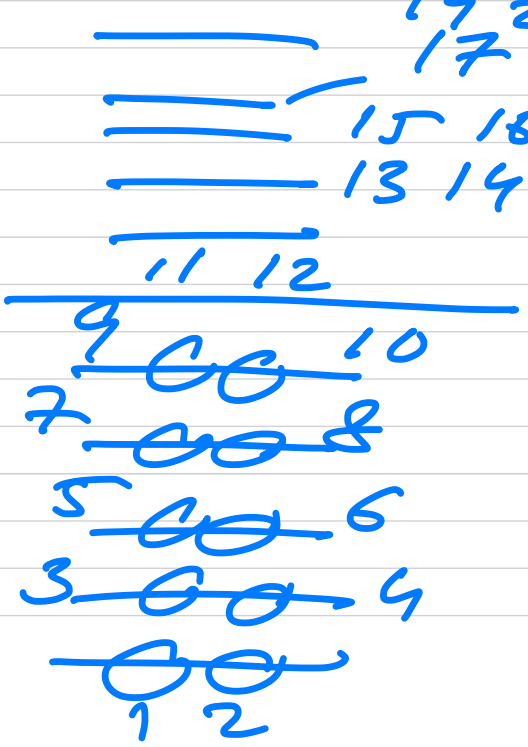
$$D_m = \begin{bmatrix} 0 & & & \\ & \frac{1}{\varepsilon_0-\varepsilon_1} & & \\ & & \frac{1}{\varepsilon_0-\varepsilon_2} & \\ & & & \ddots \\ & & & & \frac{1}{\varepsilon_0-\varepsilon_5} \end{bmatrix}$$

$$V_{m0} = \begin{bmatrix} 0 \\ -s/2 \\ -s/2 \\ -s/2 \\ -s/2 \\ 0 \end{bmatrix}$$

$V_{om} \cdot D_m \cdot V_{mo} = -g^2 \frac{7}{24}$

$N = 10$ and 10 doubly deg levels

no broken pairs (and their config) $\binom{10}{5} = 252$



all states $\binom{20}{10} \sim 200 \cdot 10^6$

$$\frac{1}{4} \sum_{\substack{ab \\ i'j'}} \frac{\bar{v}_{ij,ab}\, v_{ab,i'j'}}{\varepsilon_{i'} + \varepsilon_{j'} - \varepsilon_a - \varepsilon_b}$$

$N = 10$ and 10 doubly deg SP states $\rightarrow$ 20 SP states

a, b have 10 states

$\sum_{ab}$ implies 10^2 summands

$\sum_{i'j'}$ implies the same

$(a,b) \rightarrow A$ = number of two-particle configs

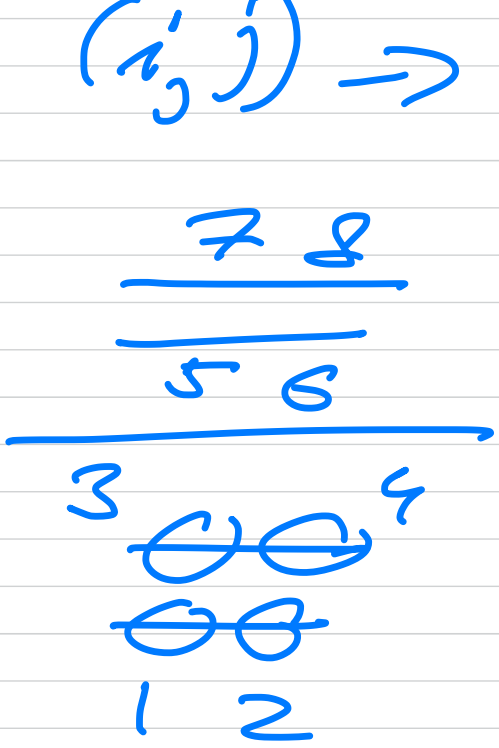
no broken pair

$N = 4$ case

$(i,j) \rightarrow I$

(a,b) : (5,6), (7,8)

(i,j) : (1,2), (3,4)



$N = 10$

$(a,b) =$ (11,12), (13,14), (15,16), (17,18), (19,20)

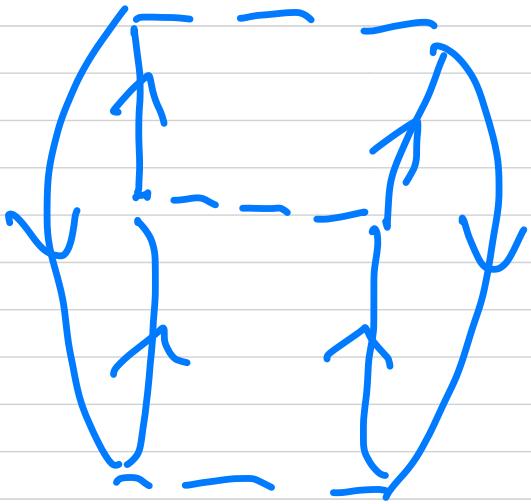
only S

$\overline{v}_{ijab} \Rightarrow V_{IA} \quad \left(\begin{matrix} w=4 \\ 2x2 \end{matrix}\right)$

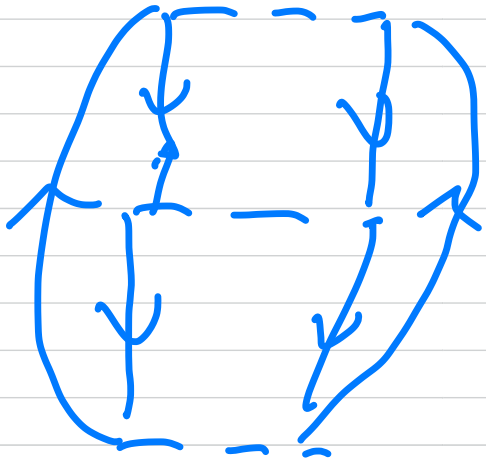
$\overline{T_{AI}}$ which is defined by $\dfrac{\overline{v}_{abij}}{\varepsilon_i + \varepsilon_j - \varepsilon_a - \varepsilon_b}$

$$\Delta E^{(2)} = \frac{1}{4} \sum_{IA} \overline{V_{IA}\, T_{AI}}$$

$$= \frac{1}{4} \overline{\mathrm{Tr}(VT)}$$



2p-1h order



2h-1h order

$$\langle \Psi_c | H_1 R_0 H_1 \underbrace{R_0 H_1}_{T} | \Psi_c \rangle$$

T T